

IVACS V-TRACE: Vehicle Trust, Route & Evidence Engine

Problem Statement OD-08: License Plate Detection and Recognition from Construction-Site CCTV Footage

Report Updated: 2026-09-18 11:26:58	Current State: All Phases Verified (100%)	Runtime AI Policy: Strict Zero LLM / Native CV
Team: CodeAvengers	Evaluation Status: DEMO READY (14/14 Tests Pass)	Git Branch: main (Synced Remote)

1. System Component Status & Subsystem Audit

Component / Subsystem	Location / Path	Operational Description	Status
Video Ingestion Engine	backend/video/	MP4, RTSP, Webcam stream abstraction with configurable frame stride	PASS
Vehicle Detection Engine	backend/detection/vehicle_detector.py	Ultralytics YOLOv8n inference filtering cars, trucks, buses, motorcycles	PASS
Plate Localization Engine	backend/detection/plate_detector.py	Bumper ROI localization, morphological gradients, and vehicle bbox association	PASS
OCR Character Engine	backend/ocr/plate_ocr.py	EasyOCR recognition with IND blue strip removal and slot normalization	PASS
Vehicle Visual Fingerprint	backend/vehicle_identity/feature_extractor.py	ResNet18 backbone yielding 512-dim L2-normalized deep visual embeddings	PASS
Identity Decision Engine	backend/vehicle_identity/identity_rules.py	Deterministic Rules A-E for plate-visual match, mismatch, and swap detection	PASS
Database Persistence Layer	backend/database/	SQLite persistence for detections, registered identities, and observation logs	PASS
Evidence Storage Engine	data/evidence/	Multi-scale image storage (full frame, vehicle crop, plate crop, annotated ROI)	PASS
REST API Application	backend/app.py	FastAPI web service with background video demo worker and static evidence mount	PASS
Automated Test Suite	tests/	14 comprehensive unit and integration tests covering all 7 pipeline areas	PASS

2. Team Member Role Allocation (CodeAvengers Team)

Member	GitHub / Email	Subsystem Ownership & Core Deliverables
Jone Kavish (Team Lead)	jonekavish-dot jonekavish@gmail.com	Backend Architecture & Identity Engine: FastAPI REST application, frame orchestrator, EasyOCR character extraction pipeline, ResNet18 visual fingerprint engine, decision rules, overall project management.
K.V. Pranesh (Member 1)	kvpranesh kvpranesh49@gmail.com	Vehicle Detection Engine: Ultralytics YOLOv8n integration, COCO multi-class vehicle filtering (car, truck, bus, motorcycle), confidence tuning, bounding box validation.
Gowshik Gunal (Member 2)	gowshikgunal22 gowshikgunal@gmail.com	License Plate Localization: Bumper ROI spatial analysis, morphological gradient edge enhancement, contour detection, vehicle-to-plate spatial association.
Dinesh Balu (Member 3)	dineshbalu7f-glitch dineshbalu7.f@gmail.com	Video & Persistence Layer: Video source abstraction (MP4 sequential stream, RTSP, Webcam), configurable frame sampling, evidence crop filesystem storage, SQLite database schema.

3. Vehicle Identity Decision Engine & Honest Detection Policy

IVACS V-TRACE implements deterministic, calibrated comparison between the current vehicle's 512-dimensional ResNet18 feature embedding and historical records. In strict accordance with hackathon evaluation guidelines, the system employs an **Honest Detection Policy**: it never claims unverified ground truth or absolute certainty, emitting qualified analytical event signals instead.

Rule	Condition / Observation	Analytical Event Signal	Operational Meaning
Rule A	Same Plate + High Similarity (≥ 0.85)	SAME_VEHICLE	Confirmed consistent identity. Observation appended to history.
Rule B	Same Plate + Low Similarity (< 0.85)	POSSIBLE_IDENTITY_MISMATCH	Vehicle visual appearance deviates from registered record for this plate. Possible plate cloning.
Rule C	Different Plate + High Similarity (≥ 0.85)	POSSIBLE_PLATE_SWAP	High visual match with a previously seen vehicle bearing a different plate. Possible plate swap anomaly.
Rule D	Plate Unreadable + High Similarity (≥ 0.85)	PLATE_UNREADABLE_VEHICLE_MATCH	Plate obscured or blurry, but vehicle visual fingerprint matches existing identity. Provides tracking continuity.
Rule E	No Historical Match Found	NEW_VEHICLE	First sighting of this vehicle. A new persistent identity record is registered in the database.

4. REST API Endpoint Specifications

Method	Endpoint Route	Description & Operational Response
GET	/api/health	Returns system health, active compute device (CPU/CUDA), and model status.
GET	/api/cameras	Lists all configured construction-site CCTV camera feeds (CAM-01 through CAM-04).
POST	/api/demo/start	Starts non-blocking background video processing on the CCTV demonstration video.
POST	/api/demo/stop	Gracefully terminates active background video processing worker.
GET	/api/demo/status	Real-time telemetry: current frame, total frames, FPS, processed count, and detection count.
GET	/api/detections	Paginated list of structured vehicle & plate detection events stored in SQLite.
GET	/api/detections/latest	Most recent vehicle detection event with plate, confidence, and crop paths.
GET	/api/vehicles	Catalog of all registered vehicle identities with visit counts and canonical plates.
GET	/api/vehicles/{id}	Detailed profile for a specific vehicle identity, timestamps, and active status.
GET	/api/vehicles/{id}/history	Historical observation timeline and camera sightings for a specific vehicle.
GET	/api/identity-events	Chronological stream of vehicle identity matching decisions and similarity scores.
GET	/api/identity-events/latest	Most recent analytical identity decision event emitted by the engine.
GET	/evidence/{filename}	Static HTTP asset access to saved evidence frames, vehicle crops, and plate crops.

5. Automated Quality Assurance & Test Verification (14/14 Passed)

Automated test suite executed via python -m pytest tests/ -v. Total execution duration: 17.80s. Pass rate: 100% (14 passed, 0 failed).

Module	Test Name	Verification Scope	Result
tests/test_identity.py	test_cosine_similarity_basics	Verifies vector math, clamping [-1.0, 1.0], and orthogonal handling	PASS
tests/test_identity.py	test_rule_new_vehicle	Validates Rule E: Registration of new vehicle identity on first sighting	PASS
tests/test_identity.py	test_rule_same_vehicle	Validates Rule A: High similarity + same plate yields SAME_VEHICLE	PASS
tests/test_identity.py	test_rule_possible_identity_mismatch	Validates Rule B: Same plate + low similarity yields POSSIBLE_IDENTITY_MISMATCH	PASS
tests/test_identity.py	test_rule_possible_plate_swap	Validates Rule C: Different plate + high similarity yields POSSIBLE_PLATE_SWAP	PASS
tests/test_identity.py	test_rule_plate_unreadable_vehicle_match	Validates Rule D: Unreadable plate + high similarity yields tracking continuity	PASS
tests/test_identity.py	test_service_identity_lifecycle	End-to-end service test: registers identity, observation, and DB persistence	PASS
tests/test_vtrace.py	test_video_source_initialization	Verifies MP4Source open, metadata retrieval, frame count, and FPS	PASS
tests/test_vtrace.py	test_invalid_video_handling	Tests robust failure handling for non-existent video files	PASS
tests/test_vtrace.py	test_frame_processing_with_no_detection	Asserts blank frames produce None without throwing runtime exceptions	PASS
tests/test_vtrace.py	test_ocr_empty_result	Ensures unreadable plate crops handle gracefully with confidence 0.0	PASS
tests/test_vtrace.py	test_valid_structured_detection_event	Validates Pydantic DetectionEvent schema and slot normalization	PASS
tests/test_vtrace.py	test_database_insertion	Verifies SQLite record insertion and query retrieval of detection events	PASS
tests/test_vtrace.py	test_evidence_image_creation	Confirms evidence images and crops are saved with correct file paths	PASS

6. Git Traceability & Remote Commit Provenance

Remote Repository: https://github.com/jonekavish-dot/VISIONX_CODEAVENGERS_OD-08.git
All commits have been authored team-member-wise and synchronized with the remote main branch.

Commit	Author	Date	Commit Message & Deliverable Scope
a68f7b3	Jone Kavish jonekavish@gmail.com	2026-09-18	feat(identity): implement Vehicle Visual Fingerprint engine, ResNet18 embeddings, decision rules A-E, and vehicle endpoints
339c7f1	Jone Kavish jonekavish@gmail.com	2026-09-18	fix(runner): add run_server.py entrypoint and update python -m uvicorn instructions
1e644db	Jone Kavish jonekavish@gmail.com	2026-09-18	feat(core, api): deliver IVACS V-TRACE foundation pipeline, EasyOCR engine, FastAPI server, tests and docs
5bab7cf	Gowshik Gunal gowshikgunal@gmail.com	2026-09-18	feat(plate-detect): implement bumper ROI license plate localization and vehicle association
79abb3d	K.V. Pranesh kvpranesh49@gmail.com	2026-09-18	feat(detection): integrate YOLOv8 vehicle detector for multi-class CCTV monitoring
9754fd7	Dinesh Balu dineshbalu7.f@gmail.com	2026-09-18	feat(video, db): implement video source abstraction and sqlite persistence layer